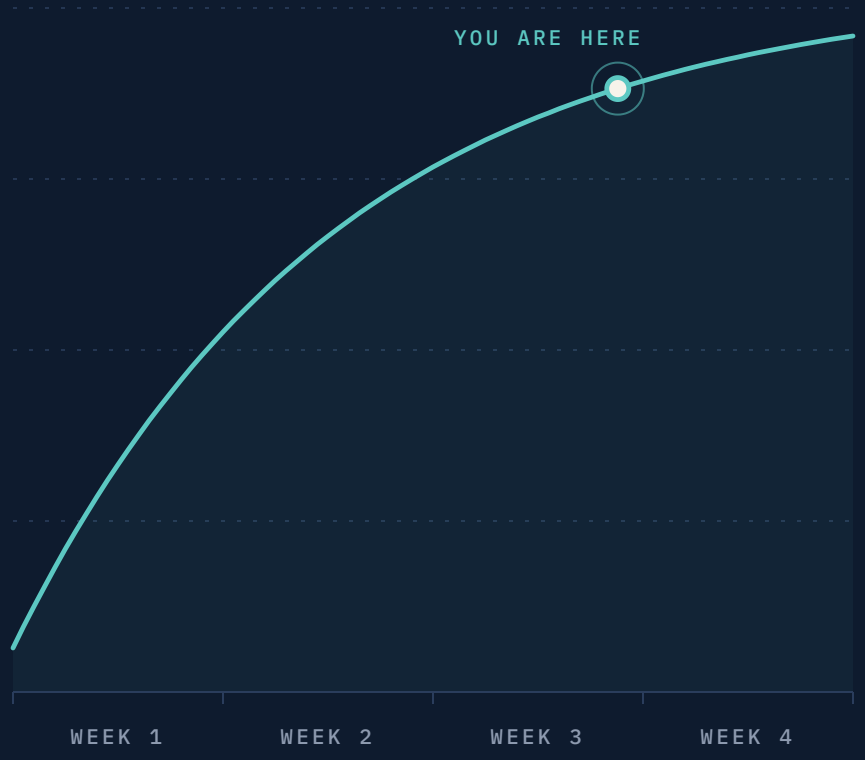
WEEK 3 · DAY 4 · OVERVIEW

Case Study

Treasury Yields



Created by **Jeff Lenamon**

ABOUT 55 MIN WITH EXERCISES / 7 SECTIONS / 7 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Problem Definition

Our investment grade bonds are quoted over Treasuries, and the desk's Treasury sheet is stale. Rebuild it from Treasury's own file, so it reruns on every download. Tell me what the file covers and what is missing, where each tenor has been by year, what the curve looked like, the 10s less 2s spread and how long it stayed below zero, and how much yields move in a day. Describe it: no why, no forecast.

ABOUT 55 MIN WITH EXERCISES / 7 SECTIONS / 7 EXERCISES

Your Route Through Day 4

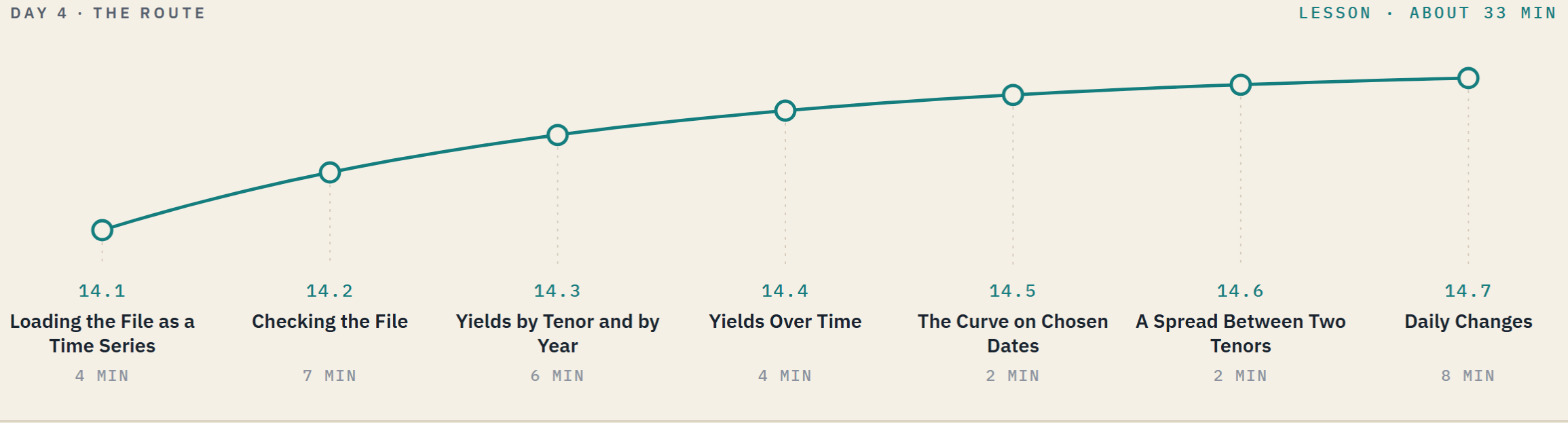
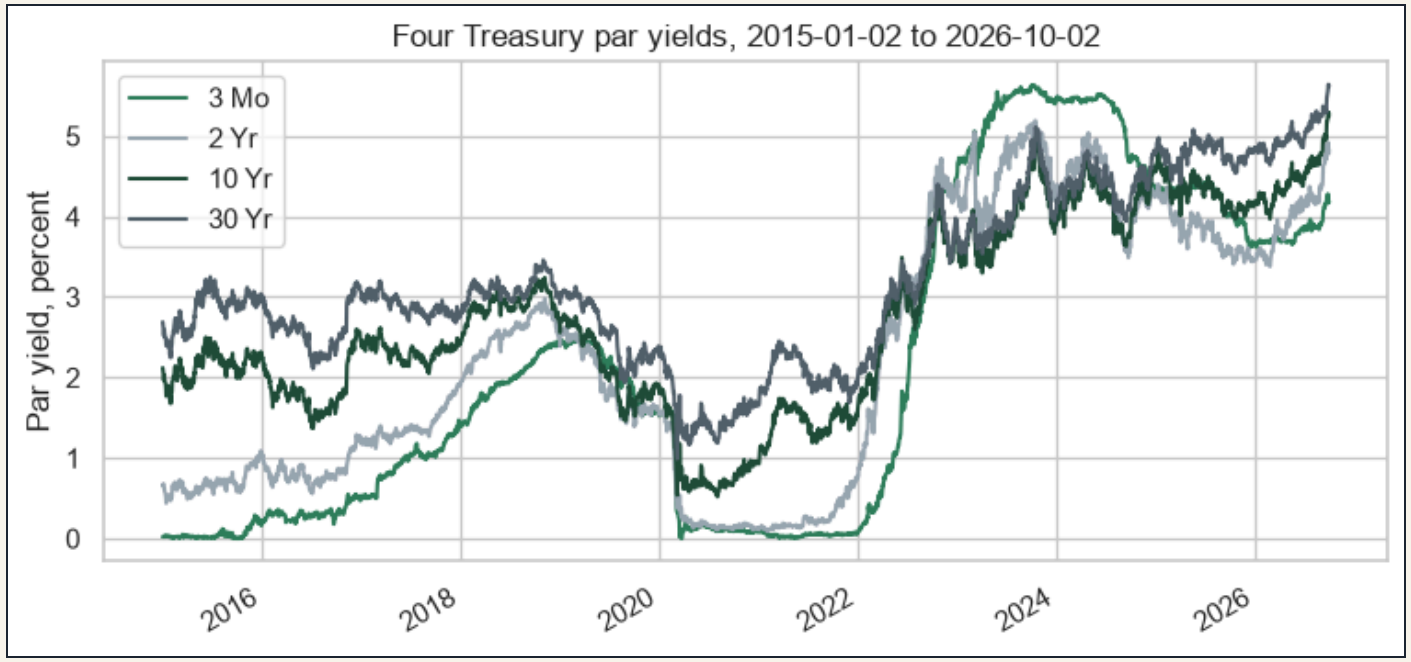


FIGURE 1 Seven sections, in order, with about how long each takes.

Time. About 33 minutes for the lesson, then about 21 for the seven exercises.

14.1 Loading the File as a Time Series



DATES AS ROW LABELS

```
print(yields.shape, type(yields.index))
```

```
(2940, 14) <class 'pandas.DatetimeIndex'>
```

EXCEL SIDE BY SIDE

Excel 365 converts the dates on opening, so `=ISNUMBER(A2)` is TRUE; column I holds the `2 Yr`, M the `10 Yr`.

A date index: pick by date, group by year, draw against time. Now check the file.

14.2 Checking the File

Two kinds of missing

Absent: a weekday with no row

date	3 Mo	2 Yr	10 Yr
2024-07-03	5.47	4.71	4.36
2024-07-04: no row at all			
2024-07-05	5.46	4.60	4.28

No cell says it is missing.

Found only against a calendar: pd.date_range(freq='B'), difference().

Blank: a row with a cell empty

date	1 Mo	2 Mo	3 Mo
2018-10-12	2.14	NaN	2.28
2018-10-15	2.17	NaN	2.31
2018-10-16	2.19	2.22	2.30

The row is there; the cell holds NaN.

Found by isna(); the first value by first_valid_index().

FIGURE 2 Two kinds of missing: a date with no row, found against a calendar, and a blank cell, found with isna.

WEEKDAYS WITH NO ROW

```
print(len(weekdays), len(yields), len(no_row))
```

```
3066 2940 126
```

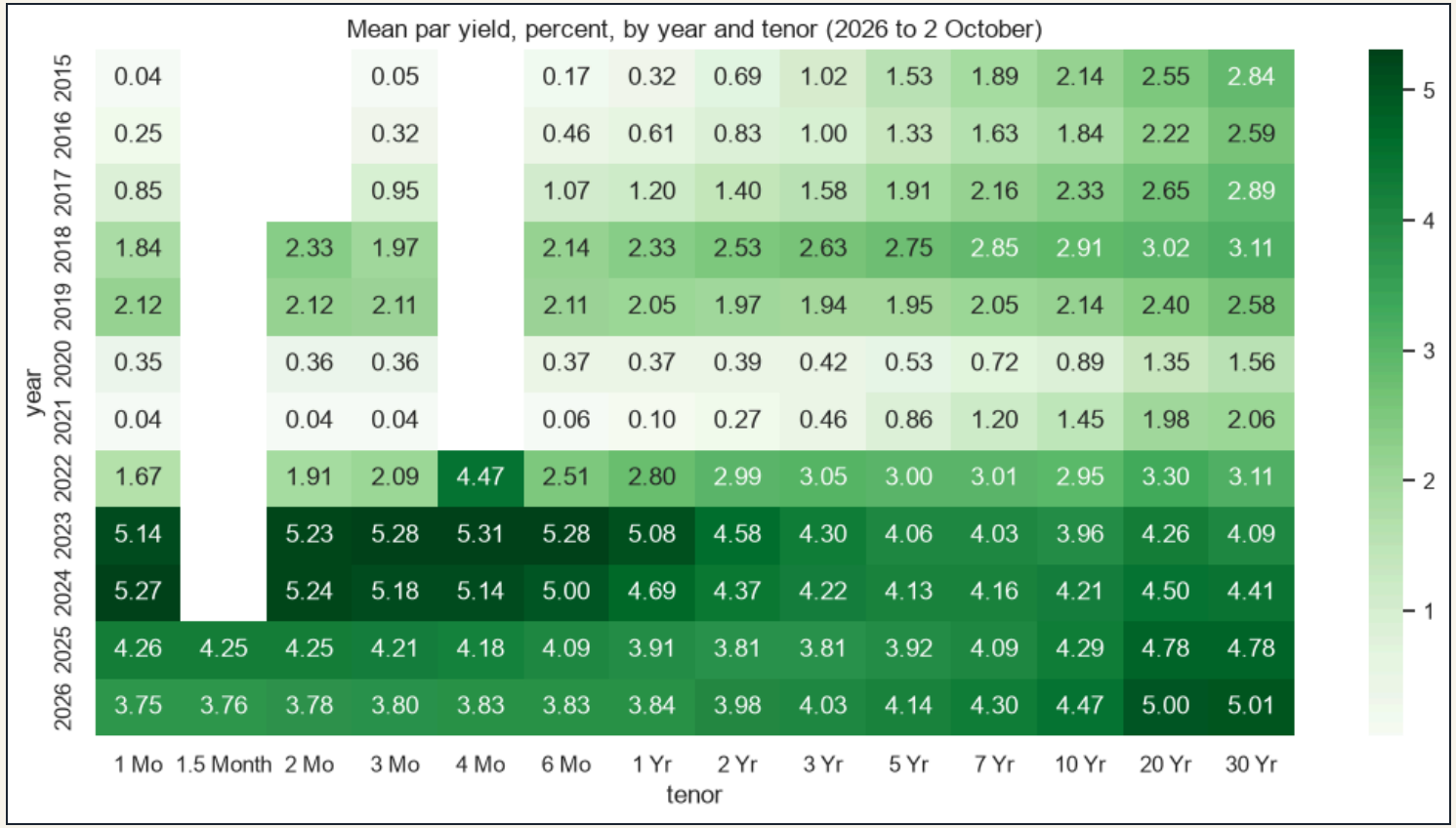
BLANK CELLS

```
print('Blank cells:', blanks.sum())
```

```
Blank cells: 5432
```

126 days absent, 5,432 cells blank, nothing filled: put `count()` beside every average.

14.3 Yields by Tenor and by Year

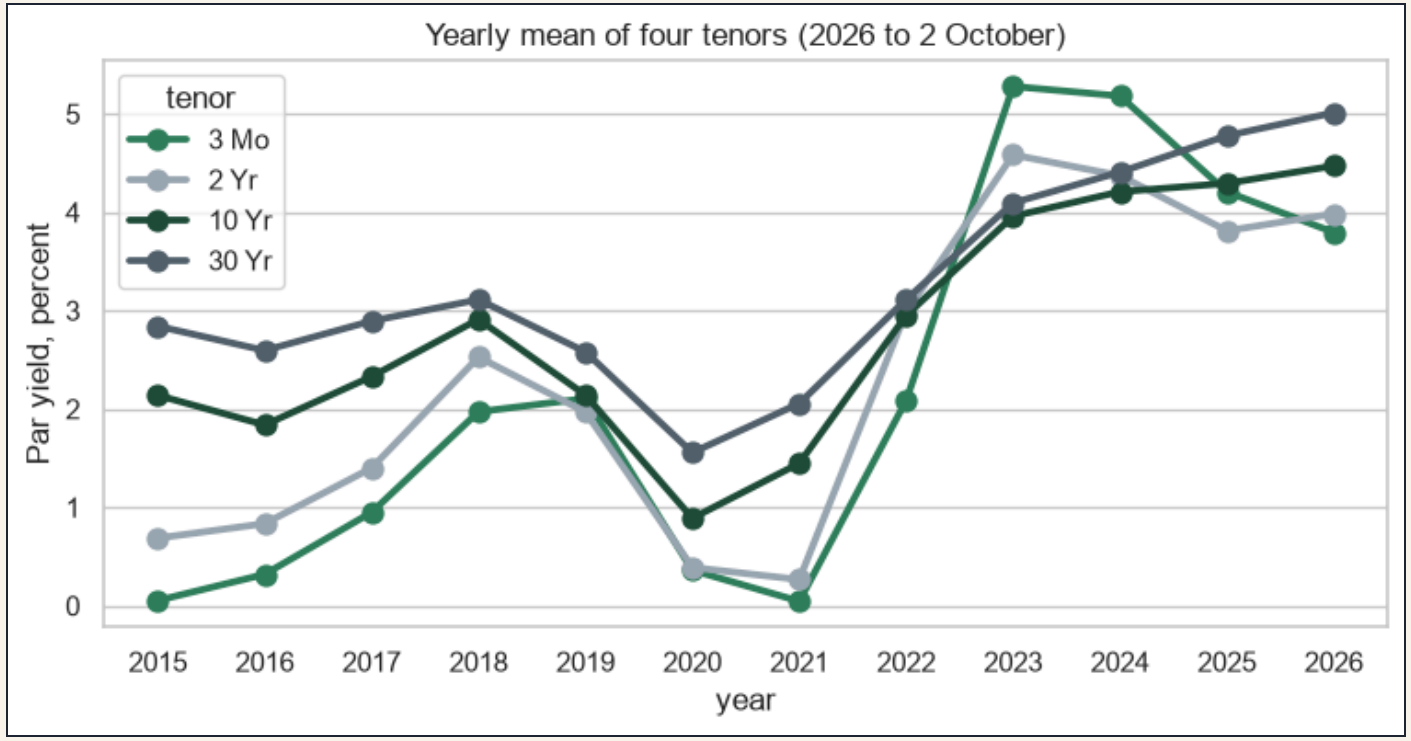


EXCEL SIDE BY SIDE

JOB	EXCEL
A year's mean	`=AVERAGEIFS(M:M, P:P, 2020)` with `=YEAR(A2)` in P; `MAXIFS`, `MINIFS` from Excel 2019
Wide to long	Power Query, Unpivot Other Columns, which leaves out null cells
Shade a table	Conditional Formatting, Color Scales

Averages by year say where each tenor sat. Next: when.

14.4 Yields Over Time



WHICH TENOR WAS HIGHEST

```
top = yields[main].idxmax(axis=1)
print(top.value_counts())
```

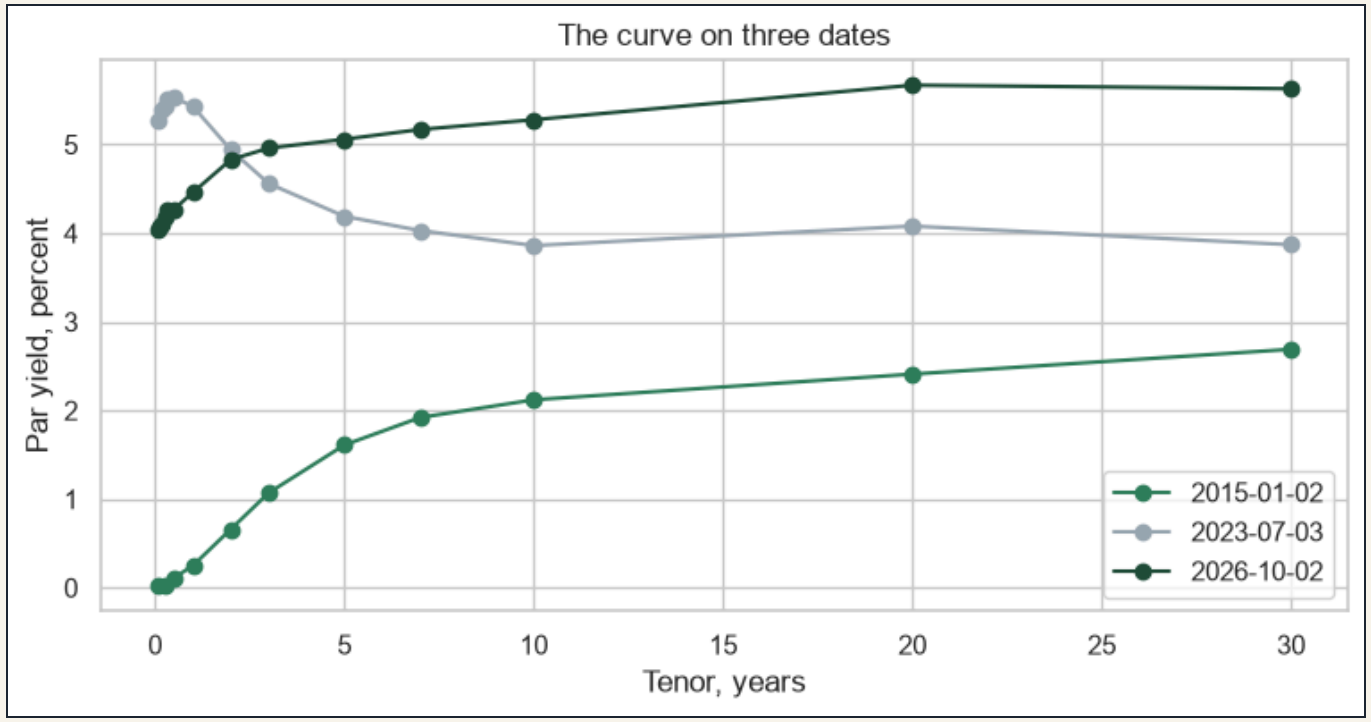
```
30 Yr    2335
3 Mo      508
```

EXCEL SIDE BY SIDE

JOB	EXCEL
Date of the high	`=XLOOKUP(MAX(M2:M2941), M2:M2941, A2:A2941)`
Month-end value	`=XLOOKUP(EOMONTH(DATE(2015,1,1),0), A2:A2941, M2:M2941, , -1)`

Say which monthly figure it is, and flag the part month.

14.5 The Curve on Chosen Dates



FIVE TENORS ON THREE DATES

DATE	1 MO	2 YR
2015-01-02	0.02	0.66
2023-07-03	5.27	4.94
2026-10-02	4.04	4.83

```
print(day.date(), len(curve), 'points')
```

```
2015-01-02 11 points
2023-07-03 13 points
2026-10-02 14 points
```

2023-07-03, chosen by rule. Next, the 10 Yr less 2 Yr on every date.

14.6 A Spread Between Two Tenors



DAYS BELOW ZERO

```
below = yields['spread_bp'] < 0
```

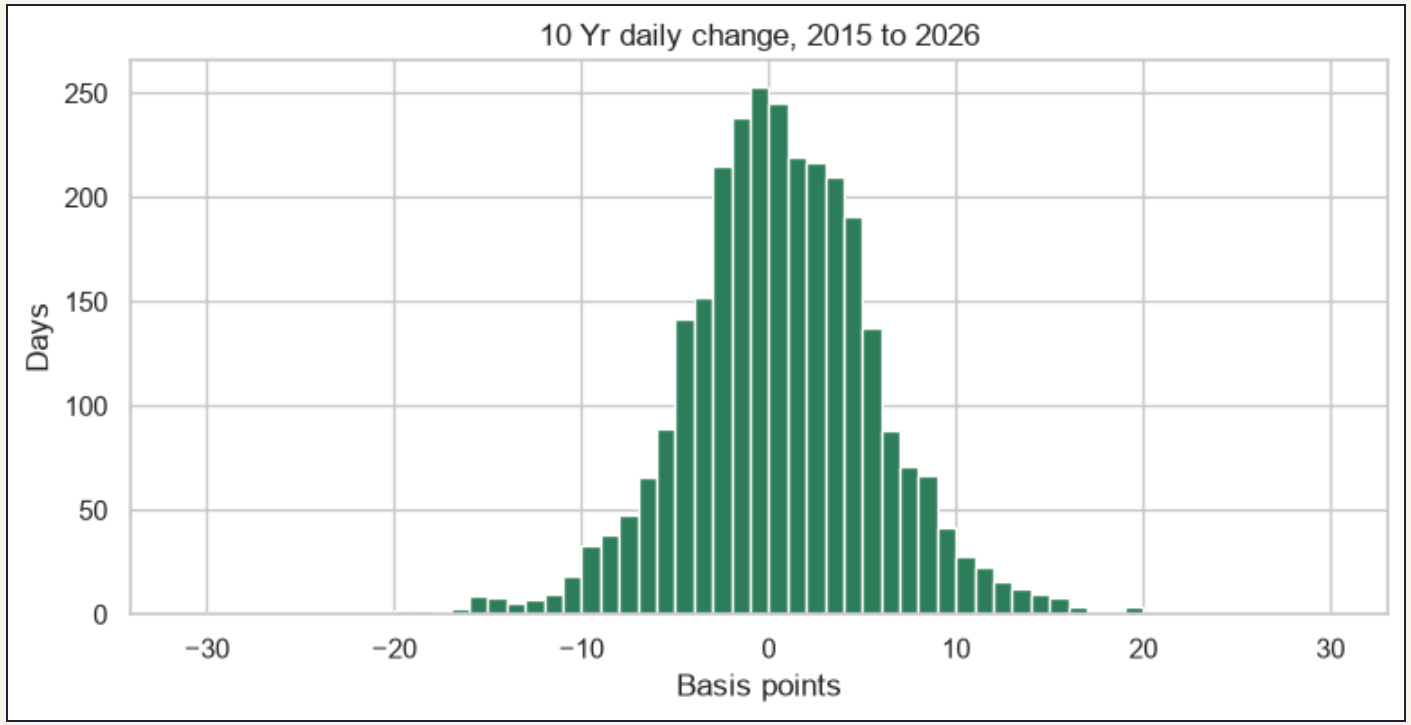
```
544 days, 18.5 percent; at zero: 7
```

EXCEL SIDE BY SIDE

JOB	EXCEL
Days below zero	`=COUNTIF(Q2:Q2941, "<0")`, with `=(M2-I2)*100` filled down Q
In one year	`=COUNTIFS(Q2:Q2941, "<0", P2:P2941, 2023)`

544 days below zero, every day of 2023. Last: how far yields move in a day.

14.7 Daily Changes



THE LARGEST MOVES

```
print(changes['10 Yr'].nlargest(3))
print(changes['10 Yr'].nsmallest(3))
```

```
date
2020-03-17    29.0
2022-06-13    28.0
2020-03-10    22.0
Name: 10 Yr, dtype: float64
date
2022-11-10   -30.0
2022-09-28   -25.0
```

5.30 bp a day over the span, 5.34 on the last date: the sheet's last line.

You Can Now

- Read a file as a time series with **`parse_dates=`** and **`index_col=`**.
- Pick dates with **`.loc`** and group by **`.index.year`**.
- Find absent weekdays with **`pd.date_range()`** and **`difference()`**.
- Reshape with **`melt()`** and draw **`relplot(kind='line')`** and **`pointplot()`**.
- Take month ends with **`resample('ME').last()`**.
- Work out changes with **`diff()`** and windows with **`rolling()`**.
- Describe a series with dates, numbers, and units, never why.

NEXT
WEEK 3 · DAY 5

Case Study: A Trade Blotter

One month of the desk's trades is checked, read a column at a time, joined to the holdings, and written up as one page.